

Monitoring the Brain

Being able to collect information from a living brain at work has revolutionized both our understanding of how the brain functions and brain medicine.

EEG

The simplest brain monitor is the electroencephalograph (EEG). It uses electrodes positioned all over the cranium to pick up an electrical field created by the activity of neurons in the cerebral cortex. The varying levels may be displayed as waves ("ordinary EEG") or colored areas (quantitative EEG, or QEEG). EEG can reveal evidence of seizure disorders, such as epilepsy, and signs of injury, inflammation, and tumors. The painless procedure is also used to assess brain activity in coma patients.

Types of EEG wave

Neighboring cells in the cortex fire in synchrony, creating wavelike changes in the intensity of the electrical field. Characteristic wave patterns (named after letters of the Greek alphabet) have been found to be closely associated with certain brain states.

WHY DOES THE BRAIN PRODUCE ELECTROMAGNETIC FIELDS?

Neurons use pulses of electric charge to transmit messages. The activity of billions of cells accumulates into a constant field.

